

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Final Term Project Submission

Purpose

The final term project should present a complete ecological niche or species distribution modeling analysis. Your report should connect a clearly defined biological question to appropriate data, modeling decisions, evaluation, and ecological interpretation.

The project will be evaluated on the quality and reproducibility of the analysis. A model with weak or conditional results can still be successful when its limitations are diagnosed and interpreted appropriately.

Before writing, identify a peer-reviewed journal that would be an appropriate outlet for the study. Review its author guidelines and examine at least two recent research articles from that journal. Pay attention to the organization, length, citation style, and level of detail used in their writing. Follow those conventions when practical, although course requirements take priority.

This final report should generally include four to six figures and one to three tables, with approximately eight total display items as a reasonable maximum. Typical figures might show the study area, occurrence records and M, environmental relationships or response curves, the primary suitability prediction, and a robustness or uncertainty comparison. Additional figures may show model transfer, environmental novelty, or an important diagnostic. Tables may summarize data sources and environmental predictors, model settings, evaluation results, or comparisons among alternative assumptions. Every figure and table should address a specific question, include a complete caption, and be referenced in the text. Additional diagnostic outputs can be placed in an appendix.

Submission components

1. A final written report in PDF format
2. A link to your private GitHub repository containing the complete analysis

Written report

Title, target journal, and abstract

Provide a descriptive title, the name of the target journal, and an abstract of approximately 200 to 300 words. Summarize the biological question, data, modeling approach, primary results, and main conclusion.

Introduction

Write the introduction as the opening of a scientific paper. It should provide the biological motivation for the study, explain the concepts needed to understand the objectives, and place the project within the relevant ecological literature. The introduction should generally contain about five connected paragraphs. A useful five-paragraph structure is:

1. **Broad biological motivation.** Introduce the larger ecological or conservation problem. Explain why the problem matters beyond the focal species or location.
2. **Relevant concepts.** Explain the ecological concepts necessary to understand the study. Define specialized terms and introduce the processes or relationships that motivate the analysis.
3. **Current knowledge and research gap.** Synthesize relevant literature. Explain what previous studies have established, what remains uncertain, and how that uncertainty leads to your study.
4. **Study context.** Provide the biological, geographic, or conservation context needed to understand the objectives. Introduce the study organism or ecological system here if it contributes directly to the motivation of the study. Otherwise, detailed natural history, occurrence data, and study-area descriptions belong in the Methods.
5. **Objectives and expectations.** State the research objective or question clearly. Present any biological hypotheses and predicted patterns. End by explaining how the analysis addresses the identified knowledge gap.

Throughout the introduction, support ecological claims with relevant references from peer-reviewed primary literature. Organize the introduction around ideas rather than summarizing one source at a time. The introduction should explain why the study is needed. Detailed descriptions of data processing, modeling algorithms, software, and evaluation procedures belong in the Methods.

Methods

The Methods should explain what you did, why you made the major analytical choices, and how another researcher could reproduce the study. Organize the section using the following subsections when applicable.

Study system and geographic scope

Describe the study organism or ecological system, study region, and time period. Include only the natural-history information needed to understand the study design. Explain any geographic, biological, or temporal boundaries placed on the analysis.

Occurrence data

Describe the data sources and access dates, geographic and temporal coverage, record-selection criteria, removal of duplicates or questionable coordinates, spatial thinning or other treatment of sampling bias, and final occurrence dataset used for modeling. Explain the reasoning behind

important filtering decisions. Do not describe occurrence records as a random sample unless the sampling design supports that claim.

Environmental data

For each predictor or dataset, report the variable name and biological rationale for including it, data source and version, units, spatial resolution and extent, time period represented, and any cropping, masking, reprojection, or resampling. Explain how you evaluated correlated predictors and why the final variables were retained.

Accessible area and background sampling

Describe how you defined the accessible area (M) and explain the biological or historical reasoning behind it. Report the geographic method used to create the boundary.

Describe the number of background locations, the sampling method, any sampling weights or bias correction, whether the same background was used across models, and any alternative definitions of M that were evaluated. Make it clear that background locations represent available environments.

Model calibration

Identify the modeling algorithm, software, package version, and important model settings.

Explain how predictors entered the model, how model complexity was controlled, any tuning or parameter selection, whether interactions or nonlinear responses were allowed, number of model runs or replicates, random seeds or other reproducibility settings. Describe model settings precisely rather than relying on terms such as “default settings.”

Model evaluation and robustness

Describe how data were divided for calibration and evaluation. Explain whether partitions were random, spatial, temporal, or geographically structured.

Report the number of partitions or repetitions, the evaluation statistics and what they measure, any threshold used to create binary predictions, how spatial clustering or sampling bias was considered, and which assumptions were varied in the robustness analysis. Identify the biological interpretation being compared across model alternatives.

Model projection

Include this subsection when projecting to another region, climate period, or future scenario.

Describe the projection region and time period, the environmental datasets used for projection, the compatibility with calibration variables, the treatment of extrapolation or clamping, and the method used to identify novel environmental conditions.

Software and reproducibility

List the software and major packages used. Provide the repository link or explain where the code and supporting data can be accessed. Identify any data that cannot be shared and provide instructions for obtaining them.

Your selected journal’s requirements may combine or rename these subsections. You may follow the journal’s organization, but your Methods must still cover all applicable information above. Results and biological interpretations should not appear in the Methods.

Results

The Results should present the evidence produced by the analysis. Organize the section around the research objectives rather than the order in which the code was run. Report weak, unexpected, and conditional results as clearly as strong results. Use the following subsections when applicable.

Data summary

Report the data that entered the final analysis, including the initial and final number of occurrence records, the geographic and temporal coverage, the number of background locations, the final predictor set, and the number of records or predictors removed during processing. Do not repeat data-processing procedures from the Methods.

Model evaluation

Report the model-evaluation results using the measures identified in the Methods. Include values for each model, partition, or replicate when appropriate. Describe how performance varied across models or validation partitions. Avoid treating a high evaluation score as evidence that the model represents the species' complete niche or predicts true occupancy.

Environmental relationships

Present the modeled relationships between occurrences and environmental predictors. This may include response curves, coefficient estimates, variable importance, environmental-space comparisons, and estimated optima or thresholds. Describe the pattern shown by the model. Reserve any detailed biological explanations for the Discussion.

Geographic predictions

Present the primary map or spatial prediction. Describe where relative suitability was higher or lower, major geographic patterns, whether known occurrences aligned with higher predicted values, and any areas with limited data or novel environments. Use the correct interpretation of the model output. Do not call relative suitability occupancy probability, abundance, or habitat quality unless the analysis supports that interpretation.

Robustness and uncertainty

Report how results changed across alternative assumptions, such as different definitions of M, alternative predictor sets, different occurrence filters, background samples, model algorithms or complexity settings, and evaluation partitions. Identify which patterns remained similar and which changed. Provide the evidence needed to classify the primary interpretation as stable, conditional, or not robust.

Model transfer or scenario projection

Include this subsection when projecting to another region, time period, or climate scenario. Report the predicted geographic changes, areas of agreement or disagreement among projections, locations with novel environmental conditions, evidence of extrapolation or clamping, and transfer-evaluation results, when available.

Figures and tables

Number figures and tables in the order they appear. Refer to each one in the text and provide a caption that explains what was analyzed, what colors or symbols represent, and any abbreviations. The Results text should identify the important pattern rather than restating every value displayed in a table or figure. Include quantitative evidence, units, sample sizes, and measures of variation when available. Reserve explanations of ecological meaning, comparisons with previous studies, management implications, and broader limitations for the Discussion.

Discussion

The Discussion should explain what the results mean in relation to the research question. A useful target length for your discussion is approximately five paragraphs following the progression of the instructions included below. Longer projects may require additional paragraphs, but the Discussion should remain organized around these following functions rather than repeating the Results.

1. **Main findings.** Begin by stating the main finding and whether it supported the biological expectations or hypotheses that you laid out in the last paragraph of the Introduction. Base this conclusion on the evidence reported in the Results, but do not repeat model scores or figure explanations here.
2. **Biological interpretation.** Interpret the modeled environmental relationships using the ecology and natural history of the study organism. Compare the findings with relevant literature and explain whether they agree with previous research. Consider biological mechanisms that could produce the observed patterns.
3. **Robustness and alternative explanations.** Discuss whether the main interpretation remained consistent across alternative modeling assumptions. Describe the conclusion as stable, conditional, or not robust. Consider reasonable alternative explanations such as dispersal limitations, species interactions, sampling bias, spatial autocorrelation, or missing predictors.
4. **Limitations and scope of inference.** Describe the most important limitations of the data and analysis and explain how they affect the conclusions. Distinguish relative environmental suitability from occupancy, abundance, habitat quality, or future colonization. Discuss extrapolation and environmental novelty when models were transferred across regions or periods.
5. **Implications and conclusions.** Explain the ecological, conservation, or management relevance of the findings without extending the conclusions beyond the evidence. Identify useful next steps or additional data needs. End with a concise statement of the study's main contribution and the most important implication.

References

Include a separate References section using a consistent citation format, preferably the format of your target journal. Ecological claims about habitat, physiology, dispersal, species interactions, or conservation concerns should be supported by relevant literature. Use peer-reviewed primary

research when possible. Include in-text citations in the sections above and complete reference entries here.

Repository requirements

Your repository should contain:

- Clean, easy to read, and annotated scripts that run from the project root
- An organized folder structure
- Documentation of occurrence filtering and data processing within the script
- Model settings and random seeds
- All figures and tables used in the report, reliably reproduced into an output folder
- A README explaining how to reproduce the analysis
- Package or software-version information
- Citations and instructions for obtaining data that cannot be shared

Do not upload restricted data or unnecessarily large temporary model files. The repository submitted through Canvas must match the final report. Acknowledge any use of AI or code-assistance tools in the README or an acknowledgments section in the written report. You remain responsible for understanding and verifying all submitted work.

Format and submission

- Due Friday, December 11 by 11:59 PM
- Up to 12 pages, excluding references and optional appendices
- PDF format
- 12-point Times New Roman
- Standard margins and double spacing
- Figures and tables embedded in the report
- Submit the PDF and private GitHub repository link through Canvas